



### THE PROBLEM

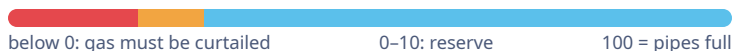
- On the record cold evening of Jan 31, 2024, Enstar told lawmakers it came "**extremely close**" to being unable to deliver gas: storage was maxed out and Hilcorp could add only about **10 million cubic feet** more.<sup>1</sup>
- Southcentral runs on **one pressurized system of pipes**. When demand outruns delivery, the gas stored in the pipes drains, pressure falls, and Enstar must cut customers: large users and power plants first, homes last.<sup>2,3</sup>
- This winter Enstar could start about **3 Bcf short**.<sup>4</sup>
- Losing pressure is slow to undo: in Rhode Island in 2019 it left **7,455 customers** without gas for about a week while every meter was shut off and relit.<sup>5</sup>
- Smart-thermostat programs elsewhere cut gas in short events, but homes reheat afterward. In ConEd's pilot that **snapback** erased about half the saving.<sup>6</sup>

### WHAT WE BUILT

A **simulation** of 25,000 Anchorage homes, not control of real thermostats. An operator console shows three charts on one clock: **modeled pipeline pressure**, outdoor temperature (forecast vs actual) and home discomfort.

- Planner**: a linear program over 24 home types and 96 hours finds the least discomfort that keeps pressure above a 10-point reserve every hour, counting the reheating that follows.
- Real forecasts**: it plans with the weather forecasts that existed at the time (archived National Blend of Models runs for Anchorage airport<sup>7</sup>) and **re-plans as each new forecast arrives**, every 6 hours, planning 0.75σ colder than forecast.
- Households**: scan the QR code, join in three taps, see your home on a live map and the system pressure, and override with one tap. The server never holds a home below **62°F**; homes needing steady heat are exempt.

### PRESSURE, IN ONE LINE



Each hour the pipes gain what wells and storage can deliver (278 MMcf/day in Feb 2024: the 268 MMcf record day plus ~10 of headroom) and lose what customers burn. Usable pipe storage: 9.54 MMcf. **(derived)** Modeled linepack margin; not psi and not Enstar telemetry.

### HOW IT'S BUILT

**Browser**  
React on Vercel;  
HiGHS LP in a Web  
Worker



**SpacetimeDB**  
runs the house  
physics every  
second



**Every screen**  
console and  
phones update  
live; same physics  
file on both sides

### RESULTS: THE FEB 2024 COLD SNAP, WITH LESS SUPPLY

Nobody was cut off in Feb 2024. We replay that weather and gas demand with less supply than it had. 25,000 homes **(assumed participation)**; numbers **(derived)** from the model, BoreaFlux re-planned on archived forecasts.

#### Near-miss

11.5 MMcf/day less supply: the most 25,000 homes can cover while holding the full reserve

**No program**    **-7** lowest pressure · 3 h below the line · 0.66 MMcf curtailed

**BoreaFlux**    **+10** lowest pressure · 0 h below · nothing curtailed

Above the curtailment line for the whole cold snap. Cost: about 4.8°F cooler across 22 setback hours (105 °F·h per home); no home below 62°F.

#### Stress

15 MMcf/day less supply **(assumed case)**

**No program**    **-31** lowest · 17 h below the line · 2.99 MMcf curtailed

**BoreaFlux**    **-16** lowest · 7 h below · 1.57 MMcf curtailed

Curtailment not removed, but about half the gas cut and 10 fewer hours below the line. Cost: 269 °F·h per home.

### WHY TRUST THE NUMBERS

| Check                                             | Published             | Ours        | Result               |
|---------------------------------------------------|-----------------------|-------------|----------------------|
| Saving kept after snapback, ConEd-like conditions | 48% <sup>6</sup>      | 46.5%       | <b>Pass (38-58%)</b> |
| Gas per home on a -20°F day                       | ~1.0 Mcf <sup>8</sup> | 1.04 Mcf    | <b>Consistent</b>    |
| Archived forecast error (RMSE), 12 h / 72 h ahead | —                     | 2.7 / 4.7°F | Why we plan colder   |

Each home's heat loss is calibrated from Enstar's 149 Mcf average home<sup>8</sup>, 30 years of Anchorage airport weather<sup>9</sup> and Alaska's gas energy content<sup>10</sup>. Every number on screen carries a sourced, derived or assumed label.

**What it doesn't do.** It doesn't close the seasonal 3 Bcf gap or control real thermostats today, and the 0.75σ forecast buffer was tuned on the Feb 2024 replay. It helps on the evenings when pressure decides who gets cut.

### SOURCES (EVERY LINK OPENED AND CHECKED, OCT 3-4, 2026)

- Anchorage Daily News, Enstar testimony to the House Energy Committee, Feb 6, 2024. [adn.com/business-economy/energy/2024/02/06/enstar-tells-lawmakers-gas-storage-was-extremely-close-to-being-depleted-during-recent-anchorage-cold-snap/](#)
- Alaska Beacon, Jul 28, 2026 (ADN reprint Jul 29). [adn.com/business-economy/energy/2026/07/29/as-winter-approaches-southcentral-alaska-utilities-are-worried-about-running-short-of-gas/](#)
- ENSTAR tariff \$1220, Interruption Program, eff. Sep 27, 2023. [enstarnaturalgas.com/wp-content/uploads/2023/10/Binder-1200.pdf](#)
- ADN editorial board, Aug 8, 2026. [adn.com/opinions/editorials/2026/08/08/editorial-the-winter-nobody-wants-to-imagine/](#)
- RI Division of Public Utilities and Carriers, Aquidneck Island investigation, Oct 30, 2019. [rip.uconn.edu/sites/g/files/xkgbur841/files/eventsactions/AI\\_Report.pdf](#)
- Con Edison, Gas Demand Response pilot report 2019/2020 (NY PSC 17-G-0606). [document.s.dps.ny.gov/public/Common/ViewDoc.aspx?DocRefId=\(7FA119BD-B04D-4378-95A3-BB26697815AC\)](#)
- Iowa Environmental Mesonet, MOS archive (NBS/NBE, PANC). [mesonet.agron.iastate.edu/api/1/docs](#)
- ADN, Energy Watch drill results, Nov 6, 2012. [adn.com/energy/article/results-are-latest-southcentral-alaska-energy-watch-drill/2012/11/06/](#)
- ACIS, Anchorage airport daily data 1996-2025. [rcc-acis.org/docs\\_webservices.html](#)
- U.S. EIA, heat content of delivered gas, Alaska, 2025. [eia.gov/dnav/ng/ng\\_cons\\_heat\\_a\\_epg0\\_vgth\\_btucf\\_a.htm](#)